

High pressure injectors

545-1380, 545-1380.1;

545-1381, 545-1381.1;

545-1382, 545-1382.1

(GB) Installation and operating instructions



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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General

1.1 Structure of the documentation

The documentation for Grundfos Alldos PVC injectors consists of:

- The information document "Operation and safety of chlorine systems":
 - General safety instructions for dealing with chlorine
- and the installation and operating instructions (this document):
 - Technical data
 - Commissioning and operating instructions
 - Device-specific safety instructions
 - Instructions on maintenance, servicing and repair

If you require further information or if any problems arise that are not discussed in sufficient detail in this document, please contact Grundfos Alldos directly for the necessary information.

1.2 Guarantee

The guarantee period lasts 12 months from the date of delivery by Grundfos Alldos.

A guarantee claim in the sense of our general conditions of sale and delivery can only be recognised if:

- The device is used according to the information in these operating instructions,
 - The device is not opened or handled incorrectly in any manner,
 - Repairs are only carried out by authorised and qualified personnel,
 - Only original spare parts are used for repairs,
 - Only components approved by Grundfos Alldos are used throughout the entire gas dosing system,
- Typical parts subject to wear are excluded from the guarantee, e.g.
- Seals, O-rings,
 - The injector nozzle, as its degree of wear depends heavily on local water quality

2. Safety information



Warning

Read the Grundfos Alldos information document "Operation and safety of chlorine systems" before commissioning.

2.1 Application of the device

Grundfos Alldos PVC injectors are used to inject Cl₂, NH₃ and SO₂ gas into water in water treatment systems within the framework of the applications described in these operating instructions.

The media that may be used with each model are specified on the type plate.



Warning

Other applications are considered as non-approved, and are not permissible. Grundfos Alldos accepts no liability for any damage resulting from incorrect use.

2.2 Obligations of the operator

The operator of the system is responsible for

- Complying with the following regulations:
 - Regulation for the prevention of accidents, chlorination of water (GUV 8.15)
 - Regulation for the prevention of accidents, pressure vessel (GUV 2.6)
 - Regulation for the prevention of accidents, gases (GUV 9.9)
 - Ordinance on workplaces (ArbStättV)
 - Ordinance on hazardous industrial substances (ArbStättV)
 - If required, compliance with additional country-specific regulations
- Training of operating personnel
- Provision of mandatory protective gear (information document "Operation and safety of chlorine systems")
- Arranging regular maintenance.

3. Technical data

Criteria for the selection of an injector are

- Suction power
- Counterpressure

- Motive water pressure (admission pressure)

Values for the injectors described here can be found in the dimensioning tables.

3.1 General data

Dosing medium	Cl ₂
Motive water pressure	16 bar
Counterpressure	7 bar
Permitted motive water temperature	+5°C to +30°C

Permitted ambient temperature	+5°C to +40°C
Gas connection	PE tube 8x11
Motive water connection - input	Flange DN25
Motive water connection - output	Flange DN32

3.2 Suction powers / connections

Injector	Suction power	Counterpressure range (bar)	Motive water connection input	Motive water connection output
545-1380	up to 500 g/h Cl ₂	16-25	Flange DN 25	Flange DN 32
545-1380.1	up to 500 g/h Cl ₂	5-16	Flange DN 25	Flange DN 32
545-1381	up to 1000 g/h Cl ₂	16-25	Flange DN 25	Flange DN 32
545-1381.1	up to 1000 g/h Cl ₂	5-16	Flange DN 25	Flange DN 32
545-1382	up to 2000 g/h Cl ₂	16-25	Flange DN 25	Flange DN 32
545-1382.1	up to 2000 g/h Cl ₂	5-16	Flange DN 25	Flange DN 32

3.3 Dimensions

3.3.1 High pressure injectors up to 2.0 kg/h Cl₂

545-138x

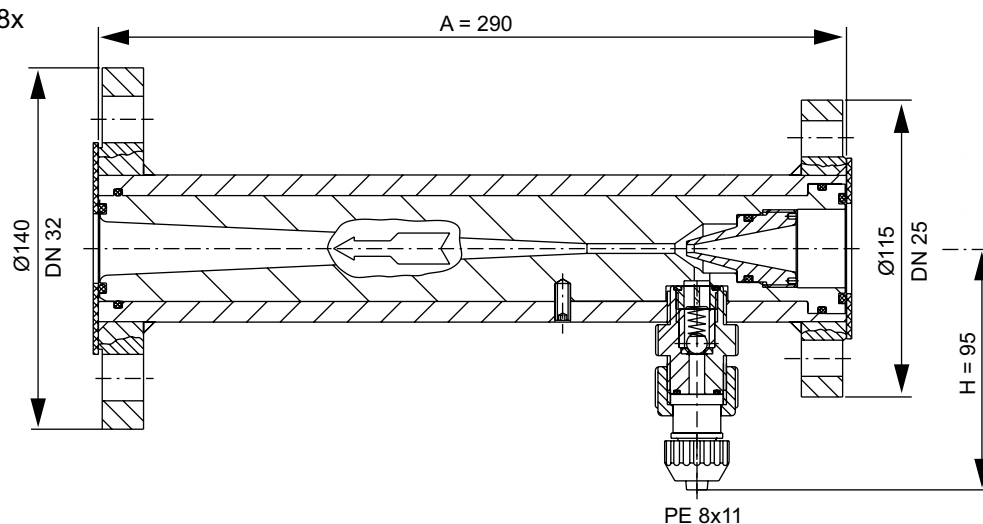


fig. 1 Dimensional drawing, 545-138x injectors

A	Overall length
H	Height

Injector	A (mm)	H (mm)
545-138x	290	95

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4. Installation

4.1 Transport and storage

- Transport the unit carefully. Never throw it.
- Store in a cool, dry location.

4.2 Unpacking

- Note when unpacking:
 - Do not allow moisture to enter the gas connection.
 - Do not allow foreign bodies to enter the connections.
- Assemble as soon as possible following unpacking.

4.3 Assembly

4.3.1 Assembly requirements

Caution Keep the solution outlet from the injector to the injection unit as short as possible.

Caution Only use sand-free, fresh water; if necessary connect a dirt trap to the water apparatus.

Caution Ensure sufficient motive water and motive water pressure (admission pressure). Fit pressure booster pump if necessary.

Note To avoid a drop in output: Observe the prescribed line cross-sections.
Observe the line cross-sections and lengths for the gas supply.

4.3.2 Fitting the injector

- Fit the injector to the wall or other suitable location using the fixing material supplied.

Observe flow direction (arrow sticker).

Caution Observe mounting position. The gas connection must point vertically downwards to ensure the ball check closes correctly.

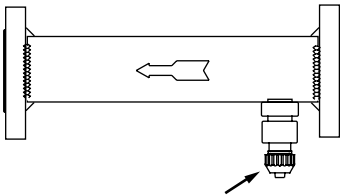


fig. 2 Mounting position, 545-138x injectors

Caution Fit and connect the injector so it is not subjected to strain.

- Connect the gas supply.
- Connect the motive water line (3),
- Connect the gas solution line (6),
- Connect the (1) gas supply,

See 6.1.2 Components.

5. Commissioning

5.1 Preparations for commissioning

Caution For commissioning, all components of the entire system must be ready for operation.
Observe the operating instructions for the other components.

5.2 Checks before commissioning

Check the tightness of the whole system before commissioning.



Warning
Only check for tightness when the entire system is ready for operation.
Risk of gas leak.

Observe the operating instructions for the other components.

5.2.1 Checking the gas solution line for tightness



Warning
System must be switched off. All tank valves must be closed.

- Disconnect the gas supply
- Open the motive water valve
- Open the injection unit isolating valve
- Switch on the motive water pump

If water escapes from between the motive water pump (3) and the injection unit (9): Leak in the line.

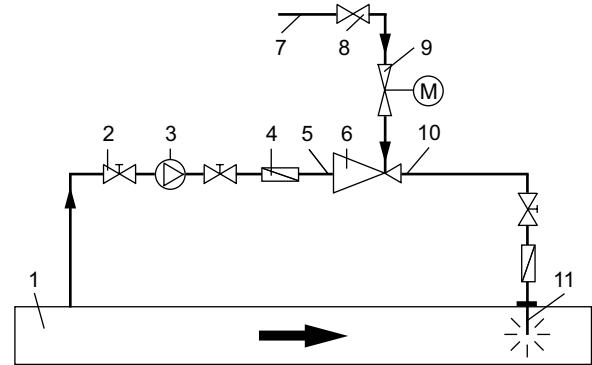


fig. 3 System diagram

1	Supply line
2	Isolating valve
3	Motive water pump
4	Check-back valve
5	Motive water input injector
6	Injector
7	Gas supply
8	Safety pressure relief valve
9	Gas dosing system
10	Gas solution line (motive water output injector)
11	Injection unit

- Switch off the motive water pump
- Close the motive water valve
- Close the injection unit isolating valve
- Repair the leak
- Repeat check for tightness.

If no water escapes: Line is watertight.

- Reconnect the gas supply

5.2.2 Check the ball check

- Disconnect the gas supply.
- Open the motive water valve.
- Open the injection unit isolating valve.
- Switch on the motive water pump.
- Briefly shut the injection unit isolating valve.
 - If water escapes from the gas connection: Faulty ball check.
- Switch off the motive water pump.
- Close the motive water valve.
- Repair the ball check (see section 7, maintenance).
- Repeat check.
 - If no water escapes: Ball check is in working order.
- Reconnect the gas supply.

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5.2.3 Checking the operation of the injector

- Disconnect the gas supply.
- Open the motive water valve.
- Open the injection unit isolating valve.
- Switch on the motive water pump.
- Place your thumb over the gas connection.

If the injector does not suck:

- Check the admission pressure (water quantity).
- Check the counterpressure.
- Check the line lengths and cross-sections.
- See section 6.3, "Possible errors"

If the error cannot be rectified:

- Contact the Grundfos Alldos service centre.

If suction can clearly be felt: Injector is in working order.

This gas is then agitated by the diffuser and is mixed with the motive water in the downstream mixing section to produce process water.

6.1.2 Components

- Nozzle (4)
 - Accelerates the motive water flow and creates a low pressure
- Diffuser with mixing section (5)
 - Mixes motive water with gas
- Body with diaphragm and ball check (2)
 - The diaphragm non-return prevents water from entering gas-conveying parts of the device
- Gas supply connection (1)
- Motive water connection - input (3)
- Gas solution line connection = motive water output (6)

6. Operation

6.1 Unit description

6.1.1 How it operates

By reducing the line cross-section with the aid of a nozzle, the motive water is accelerated and creates a low pressure. This low pressure sucks in the gas.

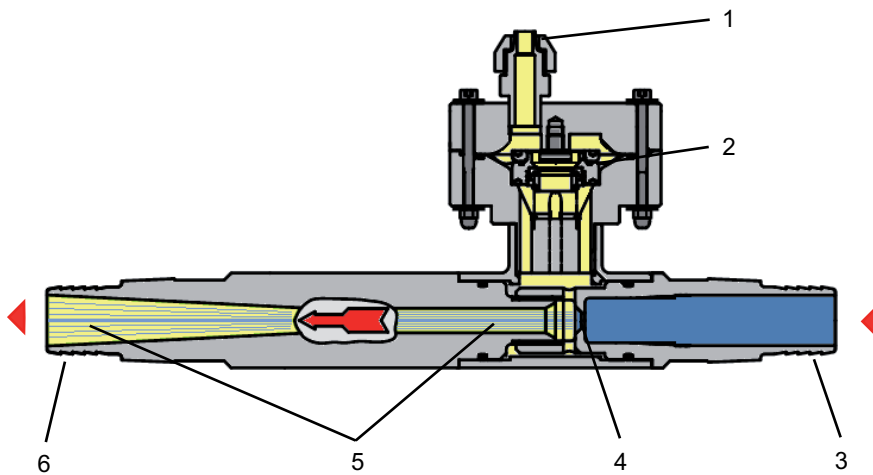


fig. 4 Injector operating diagram

6.2 Operation

Note

The device is passive. All settings must be made on other parts of the system. Observe the operating instructions for other components.

6.3 Possible errors

Error	Cause	Remedy
Maximum dosing flow is not achieved despite correctly selected injector.	Swirl unit in the nozzle damaged or not present.	Insert new swirl unit
	Counterpressure on the injector is too high (solution outlet too long or line cross-section too small).	Bring the injector closer to the injection unit, observe the line cross-sections
	Motive water pressure or quantity not correctly set.	Check the settings using the dimensioning table
	Dynamic pressure on the injector too low due to line loss.	Check length and cross-section of motive water and solution line, use the least possible number of elbows.
	Process water temperature higher than 30 °C.	Adjust process water temperature to approx. 20 - 25 °C
	The dynamic pressure created in the injector is too low due to faulty gas supply.	Replace the gas supply

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7. Maintenance

Warning



The whole system must be turned off before carrying out any cleaning or maintenance work.

Risk of gas leak.

The motive water line (input and output) must be depressurised.

Warning



Before recommissioning carry out check for tightness.

Risk of gas leak.

7.1 Intervals for cleaning and maintenance

Component	Period
Upstream dirt trap	Every 12 weeks
Nozzle	Every 12 weeks
Ball check	Every 6 months

- Additional cleaning and maintenance in the event of faults.

Note

Local circumstances (poor water quality, extreme pressure conditions) may shorten the maintenance intervals.

7.2 Cleaning the nozzle

Deposits prevent the injector from creating the required low pressure, leading to drop in output.

- Recommended cleaning agent: Hydrochloric acid



Warning

Observe the regulations when using hydrochloric acid.

Caution

Do not use sharp objects for cleaning.

- Disconnect the gas supply.



Warning

The motive water line (input and output) must be depressurised.

- Detach the injector.
- Unscrew the nozzle (pos 040, see fig.6).
- Clean the nozzle with hydrochloric acid, then rinse with clean water.
- Reassemble and fit the injector in the reverse order.

7.3 Ball check

Caution

Regular checks are necessary due to wear to the piston, seat and diaphragm(s). The checks must be carried out approx. twice a year.

7.3.1 Check the closing mechanism.

- Close the adjustment valve on the dosing system.
- Disconnect the gas supply.
- Briefly close the isolating valve behind the injector several times.
 - If no water escapes: Ball check is in working order.
 - If water escapes:
 - Replace seat (pos. 80.5, see fig.5)
- Open the isolating valve.
- Connect the gas supply.
- Set the adjustment valve on the dosing system to the required dosing flow.

7.3.2 Change the seat and ball.

- Shut down the system.
- Close the isolating valve behind the injector.
- Disconnect the gas supply.



Warning

The motive water line (input and output) must be depressurised.

- Unscrew the ball check (pos 040, see fig. 6).

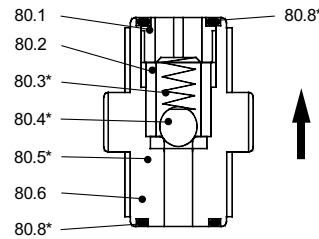


fig. 5 Change the valve seat and ball.

- Unscrew the screw part (pos 80.1, see fig.5).
- Remove the spring (pos. 80.3, see fig.5), spacer (pos. 80.2, see fig.5), ball (pos. 80.4, see fig.5) and seat (pos. 80.5, see fig.5).
- If necessary: Replace the ball, seat and spring.
- Reassemble all parts in the reverse order.
 - Screw in the screw part as far as it will go.
- Screw the ball check into the injector.
- Connect the gas supply.

8. Spare parts

We recommend that you keep some spare parts in stock to ensure that worn parts can be replaced quickly when necessary. Injection unit

8.1 Injectors up to 2.0 kg/h Cl₂

545-138x

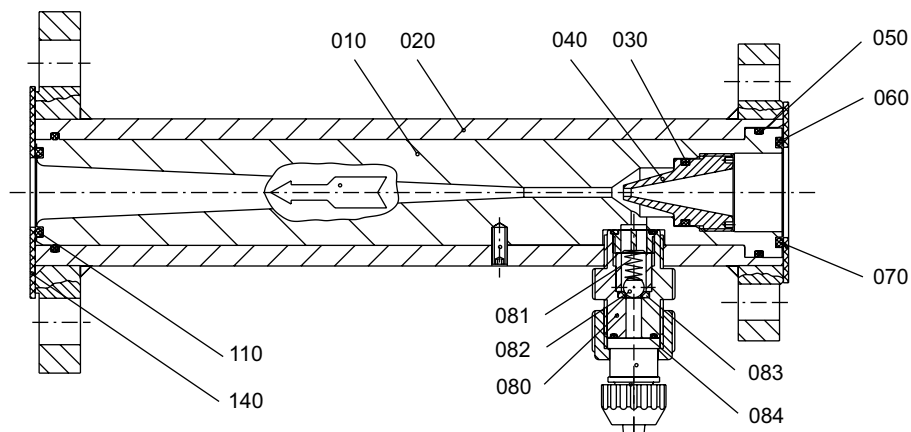


fig. 6 Assembly diagram, 545-138x injectors

The spare parts set for 545-1380 / -1381 / -1382 injectors has order no. 553-1309 / 96688796, for 545-1380.1 / -1381.1 / -1382.1 injectors the order no. is 553-1309.1 / 96688797. Each contains the following parts:

No.	Designation	Pieces
030	O-ring (20x26)	1
050	O-ring (40x46)	2
060	Flat gasket	1
070	O-ring (28x35)	1
081	Spring	1
082	Balls	1
083	Valve seat	1
084	O-ring (36x43)	2
110	O-ring (13x18)	1
140	Flat gasket	1

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9. Dimensioning tables

Criteria for the selection of an injector are

- Suction power
- Counterpressure
- Motive water pressure (admission pressure)

When dimensioning an injector, the required suction power and counterpressure is generally specified.

Other parameters such as motive water pressure and water quantity per hour can be found in the dimensioning tables.

9.1 Injectors up to 500 g/h Cl₂

Counterpressure (bar)	Motive water pressure (bar)	Water quantity (m ³ /h)
5.5	10	1.10
7	12	1.20
8	14	1.30
9.5	16	1.40
11	18	1.50
12	20	1.60
13	22	1.65
14.5	24	1.70

9.2 Injectors up to 1000 g/h Cl₂

Counterpressure (bar)	Motive water pressure (bar)	Water quantity (m ³ /h)
5.5	10	1.55
7	12	1.70
8	14	1.85
9.5	16	1.95
11	18	2.1
12	20	2.2
13	22	2.3
14.5	24	2.4

9.3 Injectors up to 2000 g/h Cl₂

Counterpressure (bar)	Motive water pressure (bar)	Water quantity (m ³ /h)
5.5	10	2.8
7	12	3.0
8	14	3.2
9.5	16	3.4
11	18	3.6
12	20	3.8
13	22	4.0
14.5	24	4.2
16	26	4.4
17	28	4.6
18.5	30	4.8
20	32	4.9
21	34	5.1
22.5	36	5.2
23.5	38	5.3
25	40	5.5

Example:

See table 9.1, injectors up to 500 g/h Cl₂.

To achieve a suction power of 500 g/h Cl₂ with a counterpressure of 13 bar, a motive water pressure (admission pressure) of 22 bar and a water quantity of 1.65 m³/h is required.

Note

All data refers to drinking water quality motive water.

Counterpressure (bar)	Motive water pressure (bar)	Water quantity (m ³ /h)
16	26	1.80
17	28	1.85
18.5	30	1.90
20	32	1.95
21	34	2.00
22.5	36	2.10
23.5	38	2.15
25	40	2.20

Counterpressure (bar)	Motive water pressure (bar)	Water quantity (m ³ /h)
16	26	2.5
17	28	2.6
18.5	30	2.7
20	32	2.8
21	34	2.9
22.5	36	3.0
23.5	38	3.1
25	40	3.2

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